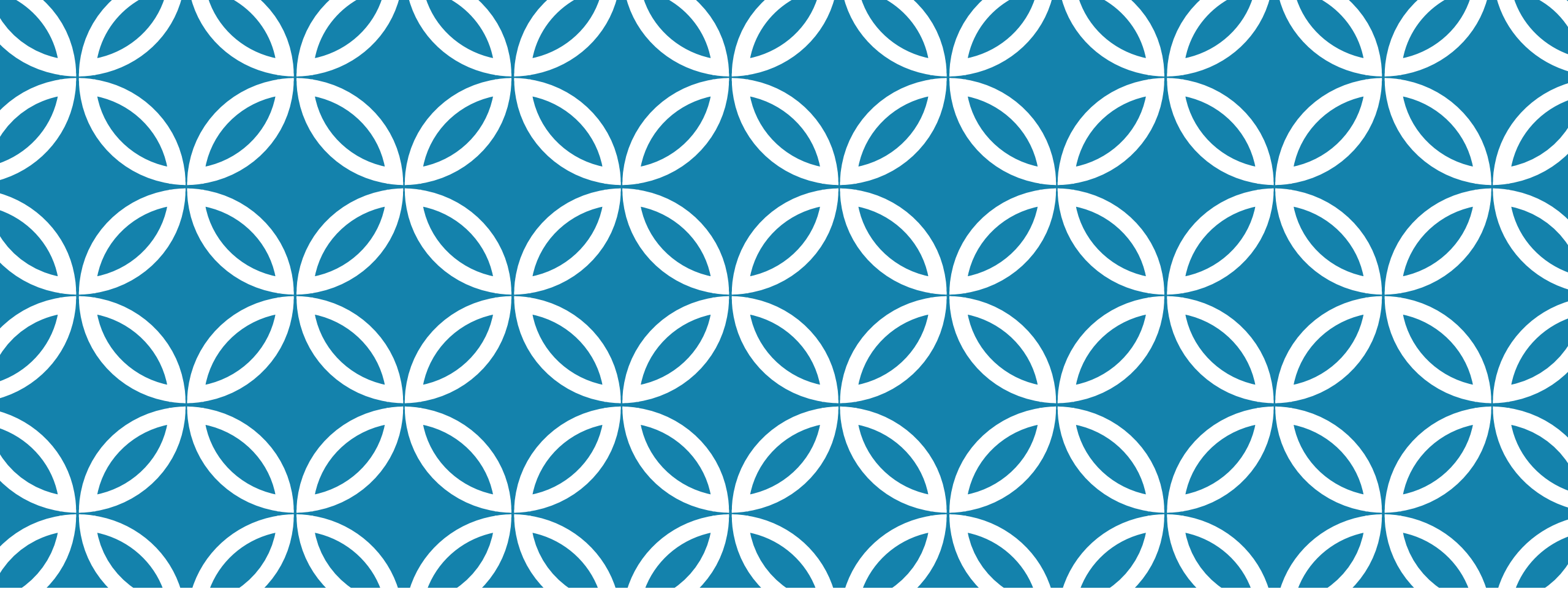




1. Sets and Relations

Logic in Computer Science

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Sets

x is an object, A and B are 2 sets

$x \in A$: x is a member of A

$x \notin A$: x is not a member of A

$B \subseteq A$: B is a subset of A

$B \subset A$: B is a valid subset of A

$P(A)$: the power set of A , that is,
the set of subsets of A

$\emptyset$ is the empty set

$1 \in \{1,2,3,4\}$: 1 is a member of $\{1,2,3,4\}$

$5 \notin \{1,2,3,4\}$: 5 is not a member of $\{1,2,3,4\}$

$\{1,4\} \subseteq \{1,2,3,4\}$: $\{1,4\}$ is a subset of $\{1,2,3,4\}$

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$\{1,4\} \subset \{1,2,3,4\}$: $\{1,4\}$ is a **valid subset** of $\{1,2,3,4\}$

$\{1,2,4\} \subset \{1,2,3,4\}$: $\{1,2,4\}$ is a **valid subset** of $\{1,2,3,4\}$

$P(\{1,2,3,4\})$: {
 $\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{3,4\},$
 $\{1,2,3\}, \{1,2,4\}, \{1,3,4\}, \{2,3,4\}, \{1,2,3,4\}, \{2,4\}$
}

Something is missing from the power set.

What is it?

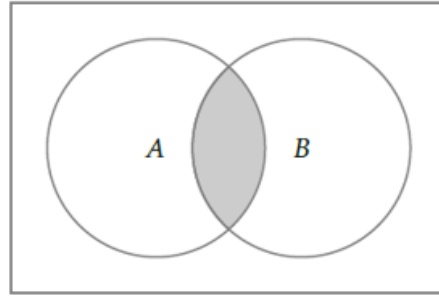
Set operations

Intersections of sets

Denoted by $A \cap B$

$x \in A \cap B$ if and only if $x \in A$ and $x \in B$

$$A \cap B$$

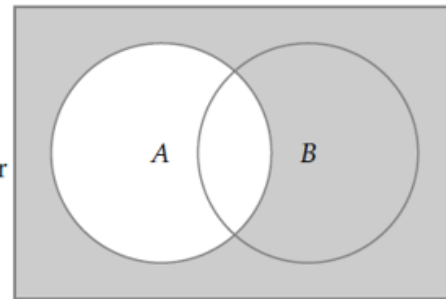


Complement of a set

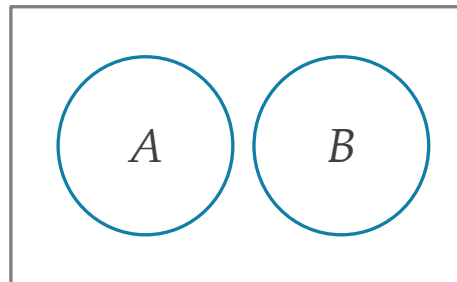
Denoted by $\overline{A}$

$x \in \overline{A}$ if and only if $x \notin A$ (of course x is a member of our universe).

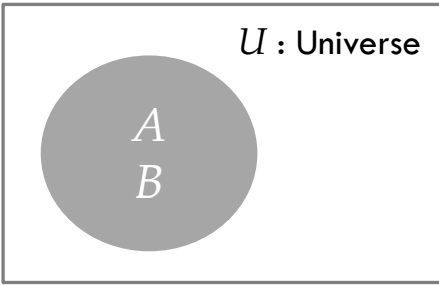
$$x \in \overline{A}$$



Disjoint sets



$$A = B$$

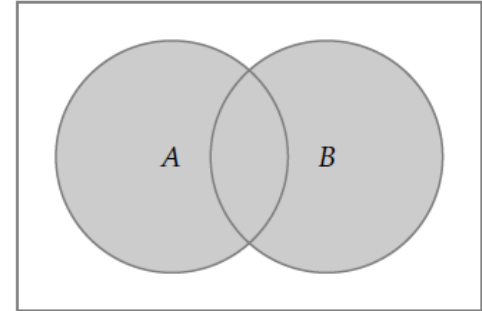


Union of sets

Denoted by $A \cup B$

$x \in A \cup B$ if and only if $x \in A$ or $x \in B$

$$A \cup B$$

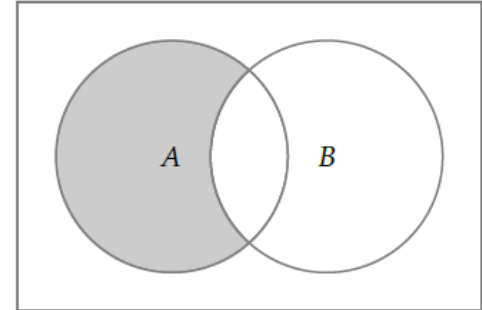


Difference between sets

Denoted by $A \setminus B$

$x \in A \setminus B$ if and only if $x \in A$ and $x \notin B$

$$A \setminus B$$

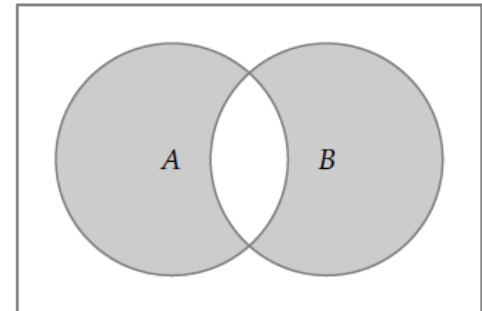


Symmetric difference of sets (also known as XOR)

Denoted by $A \Delta B$

$x \in A \Delta B$ if and only if,
($x \in A$ and $x \notin B$) or ($x \notin A$ and $x \in B$)

$$A \Delta B$$



Relations

$A \times B$: is the Cartesian product of A and B

(read: A cross B , or A times B) where A and B are nonempty sets

$= \{\langle x, y \rangle \mid x \in A, y \in B\}$, so $\langle x, y \rangle$ is an ordered pair.

n -ary Cartesian product:

If we have $n (n \geq 3)$ nonempty sets $S_1, S_2, \dots, S_n$, then the Cartesian product of the sets $S_1, S_2, \dots, S_n$ is the following set:

$$S_1 \times S_2 \times \dots \times S_n = \{\langle x_1, x_2, \dots, x_n \rangle \mid x_1 \in S_1, x_2 \in S_2, \dots, x_n \in S_n\},$$

where $\langle x_1, x_2, \dots, x_n \rangle$ is an ordered n -tuple.

Relations

Any subset of an n -ary Cartesian product is called a **relation**.

If $R \subset A \times A \times \cdots \times A$ then R is called as an **n -ary relation over A** .

If $n = 2$, this is a **binary relation**.

Denotations: $(x, y) \in R$ or $R(x, y)$

Properties of Binary Relations

With a binary relation, we will define relations between elements of two sets.

When $(h_1, h_2) \in R$ (h_1 is related to h_2) then we use the notation: $h_1 R h_2$.

The $R \subseteq H \times H$ is a binary relation on H .

A binary relation is

- a reflexive relation, if $h R h$ for all $h \in H$,
- an anti-reflexive relation, if there is no $h \in H$ where $h R h$,
- a symmetric relation, if (for all $h_1, h_2 \in H$) when $h_1 R h_2$ then $h_2 R h_1$,
- an asymmetric relation if $h_1 R h_2$ then $h_2 \not R h_1$ for all $h_1, h_2 \in H$
- an anti-symmetric relation: if there is no $h_1 R h_2$ such that $h_2 R h_1$, unless $h_1 = h_2$,
- a transitive relation, if (for all $h_1, h_2, h_3 \in H$) when $h_1 R h_2$ and $h_2 R h_3$ then $h_1 R h_3$

Binary relations

Not every relation which is not reflexive is anti-reflexive.

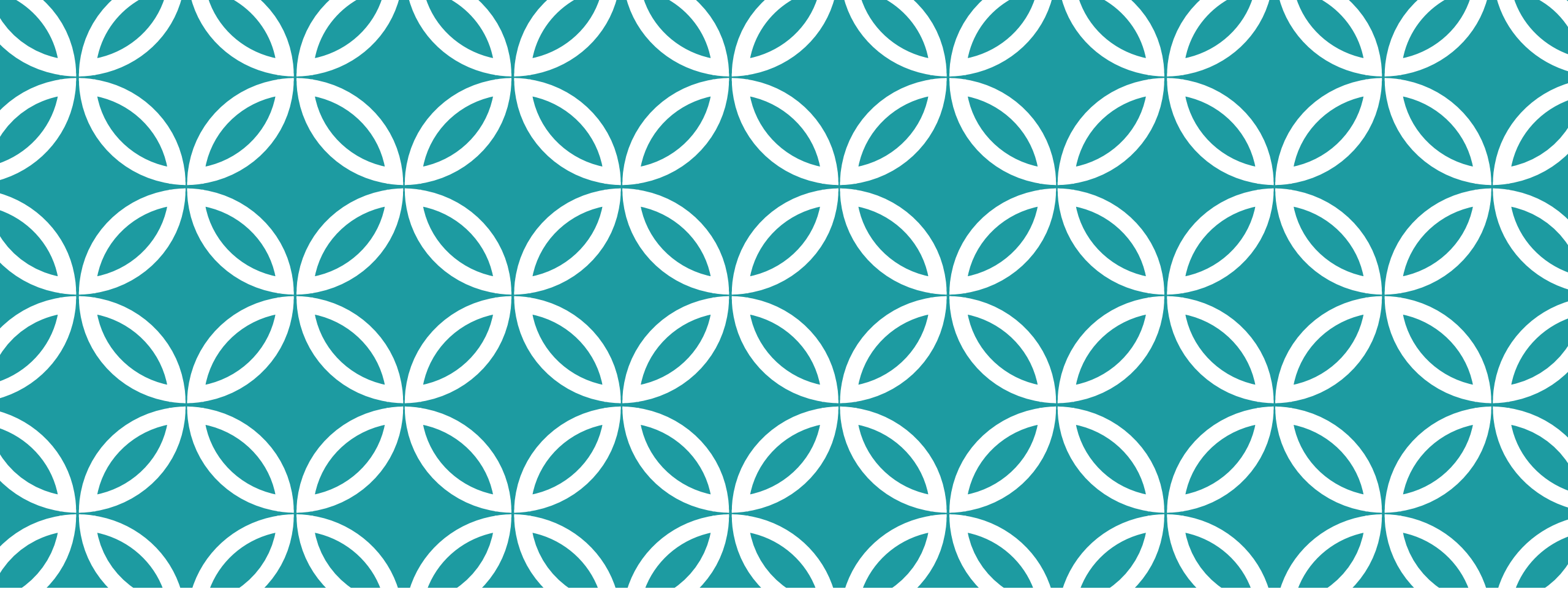
We can define relations where some elements are related to themselves but others are not (that is, neither all nor none are).

Example

The binary relation

“the product of x and y is even”

- **reflexive** on the set of **even** numbers,
- **anti-reflexive** on the set of **odd** numbers, and
- **neither reflexive nor anti-reflexive** on the set of **natural** numbers

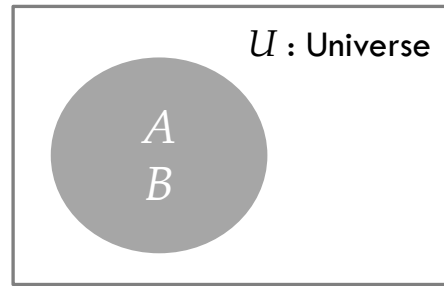


Exercises



Sets

$$A = B$$

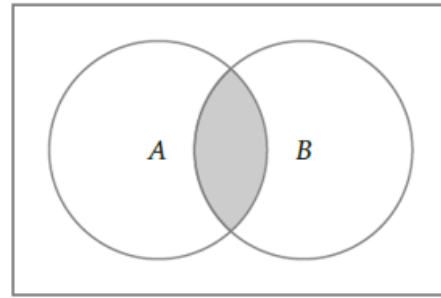


Intersections of sets

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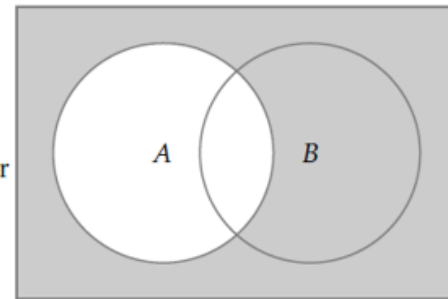


Complement of a set

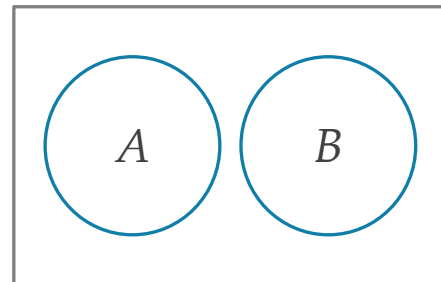
Denoted by $\overline{A}$

$x \in \overline{A}$ if and only if $x \notin A$ (of course x is a member of our universe).

$$x \in \overline{A}$$



Disjoint sets

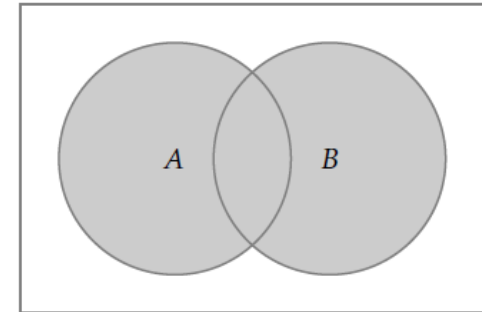


Union of sets

Denoted by $A \cup B$

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$$A \cup B$$

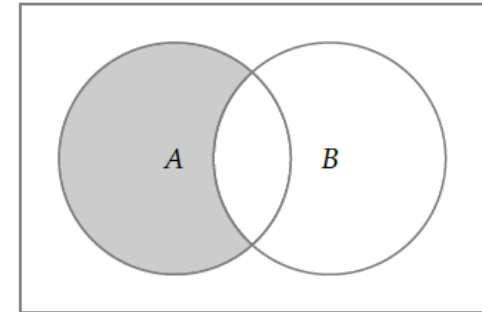


Difference between sets

Denoted by $A \setminus B$

$x \in A \setminus B$ if and only if $x \in A$ and $x \notin B$

$$A \setminus B$$

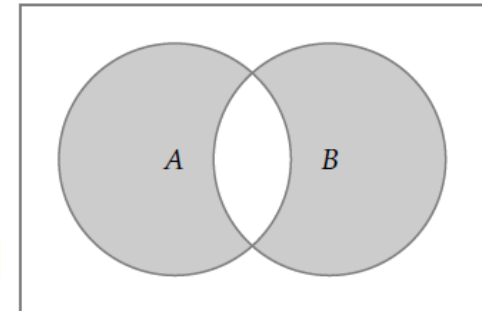


Symmetric difference of sets (also known as XOR)

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$$A \Delta B$$



Exercises

- Let A is the set of positive even numbers are less than 10, and B is the set of prime numbers are less than 10! Give the following sets with its members!
 - $A \cup B$,
 - $A \cap B$,
 - $A \setminus B$, and
 - $B \setminus A$!
- Give the Cartesian product of the set of one digit square numbers (set A) and the set of prime numbers less than 6 (set B)!
- Give the power set of set $A = \{x, y, z\}$!
- Which properties are valid for any set A , B and C ?
A. $A \setminus B = B \setminus A$ B. $A \cup (A \cap B) = B$ C. $A \setminus A = A$
D. $(A \setminus B) \cap (B \setminus A) = \emptyset$ E. $A \Delta (B \Delta C) = (A \Delta B) \Delta C$
- Which properties are valid for any set A , B and C ?
A. $B \subseteq (A \cup B)$ B. $A \cap (B \cup C) = (A \cup B) \cap C$ C. $A \times B = B \times A$ D. $A \cup (A \cap B) = A$
E. $A \Delta \emptyset = \emptyset$

Exercises

7. Which properties are valid for any set A , B and C ? A. $A \cup \emptyset = \emptyset$ B. $A \Delta B = (A \cup B) \setminus (A \cap B)$
C. $A \cap (A \cup B) = A$ D. $\emptyset \setminus A = \emptyset$ E. $A \Delta A = A$
8. Which properties are valid for any set A , B and C ? A. $A \cap B = B \cap A$ B. $A \cup A = A$ C. $A \cap B \subseteq B$
D. $(A \setminus B) \cup B = A \cup B$ E. $A \times (B \times C) = (A \times B) \times C$
9. Which properties are valid for any set A , B and C ?
A. $A \cup B = B \cup A$ B. $A \subseteq (A \cup B)$ C. $A \cap A = \emptyset$ D. $A \cap B \subseteq A$ E. $(A \setminus B) \cap B = A \setminus B$
10. Which properties are valid for any set A , B and C ? A. $A \setminus \emptyset = A$ B. $A \cap \emptyset = A$ C. $A \cap (B \cap C) = (A \cap B) \cap C$
D. $(A \setminus B) \cap A = \emptyset$ E. $A \Delta (A \Delta B) = A$
11. Which properties are valid for any set A , B and C ? A. $A \Delta B = B \Delta A$ B. $A \Delta (A \Delta B) = B$ C. $\emptyset \times A = \emptyset$
D. $(A \cup B) \times C = (A \times C) \cup (B \times C)$ E. $A \times \emptyset = A$